
VENTURE CAPITALISTS'

INVOLVEMENT IN

THEIR INVESTMENTS:

EXTENT AND PERFORMANCE

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EXECUTIVE SUMMARY

Venture capitalists responded to a questionnaire that asked them to identify their degree of involvement in a number of activities for a funded venture as well as other characteristics of the venture, including its performance. The three-page questionnaire was distributed to a sample of 350 venture capitalists during December 1986 and February 1987. In all, 62 questionnaires (18%) were completed and returned.

The results indicated that venture capitalists were involved most—compared to the entrepreneur—in the financial aspects of the venture. The activity that had the highest degree of involvement was serving as a sounding board to the entrepreneur. The lowest degree of involvement occurred in those activities concerning the ongoing operations. Factor analysis on involvement patterns in the venture activities identified four distinct areas of involvement: development and operations, management selection, personnel management, and financial participation.

The venture capitalists indicated that if they could change their degree of involvement, overall they would have done so only slightly. It was evident, however, that they would have increased their involvement in those activities requiring a minimal time commitment, such as formulating business strategy or marketing plans, or serving as a sounding board to the entrepreneur. It was evident that they would have decreased their involvement in activities that required substantial time commitment, such as developing production or service techniques, selecting vendors and equipment, or soliciting customers or distributors.

Perhaps the most important result was the identification of three distinct levels of involvement adopted by venture capitalists: 1) Laissez Faire involvement, in which the venture capitalists exhibited limited involvement; 2) Moderate involvement, in which venture capitalists exhibited moderate involvement; and 3) Close Tracker involvement, in which venture capitalists exhibited more involvement than the entrepreneur in a majority of the identified activities. Because tests regarding the venture firms, the products or services in relation to the market, and management team characteristics did not significantly explain why the three distinct types of venture capitalist involvement emerged, it appears that venture capitalists exhibited different involvement levels solely because they elected to do so. Tests also indicated that the difference in the performance level of the ventures among the three groups was statistically insignificant.

Regression analyses indicated that for each of the three types of involvement, involvement in various activities had different correlations with performance. Among Laissez Faire involvement ventures, developing the professional support group had a positive correlation with venture performance. Among Moderate involvement ventures, monitoring operations had a positive correlation with venture performance while involvement at the strategic level and in searching for management candidates exhibited negative correlations. Finally, among Close Tracker involvement ventures, negotiating employment terms with management had a positive correlation with performance, while searching for management candidates exhibited a negative correlation. It is interesting that searching for management candidates in both the Moderate and Close Tracker ventures had a negative correlation with venture performance.

These results are important because they show that depending on the involvement types selected by venture capitalists, different involvement strategies in the various activities should be more suitable. If venture capitalists recognize this they can adopt more appropriate strategies, which may lead to more successful ventures.

INTRODUCTION

A study by Gorman and Sahlman (1986) indicates that venture capitalists do not spend an inordinate amount of time directly involved with the management of their portfolio companies, intervening only cursorily in the day to day operations of those companies. According to their study the amount of involvement depends on whether the venture capitalist is a lead/early stage or a non-lead/late stage investor. A typical early stage investment receives an average of approximately two hours of direct attention per week from its lead venture capitalist while a late-stage investment receives approximately three-quarters of an hour per week from a non-lead investor. Lead investors visit their companies approximately one and a half times per month and stay for approximately five hours per visit. A non-lead investor involved at the early stage visits half as often and stays for approximately two-thirds as long. Because lead investing venture capitalists play more active roles in their portfolio companies, this study focuses solely on lead investor involvement in their selected ventures.

Several studies have indicated that, in addition to providing capital, venture capitalists perform many other roles in their portfolio companies (Gorman and Sahlman 1986; Timmons and Bygrave 1986; Timmons 1985; Davis and Stetson 1987). However, none of these studies has empirically examined how the venture capitalists' involvement correlates with the venture performance. This study attempts to identify any correlations between venture capitalist involvement with venture performance.

METHODOLOGY

Some of the ways that venture capitalists are directly involved in the venture are: 1) assistance in finding and selecting key management team personnel; 2) solicitation of essential suppliers and customers; 3) strategic planning; 4) assistance in obtaining additional financing; 5) operational planning; and 6) replacement of management personnel when appropriate. The questionnaire was designed to consider the venture capitalists' involvement that had been identified in these earlier studies. Venture capitalists were asked to indicate, for a specific venture, the extent of their involvement in each one of 20 activities, as well as the organization level of the venture capitalist representatives that intervened. In addition, the venture capitalists were asked to respond to several questions regarding the selected venture, including the venture capital firm's characteristics and the venture's product or service in relation to its market. Finally, the venture capitalists were asked to evaluate the performance of the venture on which they reported. A sample of the questionnaire is provided in the Appendix.

TABLE 1 Profile of Respondents

	Mean	Standard Deviation	Maximum	Minimum
Amount of capital under management (millions)	\$93.5	\$89.8	\$400.0	\$1.5
Years typical investments are held	5.60	1.23	10.00	3.00
Years firm has been in venture capital business	10.95	6.82	26.00	2.00
Number of senior-level professionals at firm	4.54	2.46	12.00	1.00
Number of mid-level professionals at firm	2.83	2.86	12.00	0.00

The targeted venture capitalists were selected from a combination of data bases from the Sol C. Snider Entrepreneurial Center of the Wharton School and Venture Economics, Inc.

Statistics of Venture Capital Firms Responding

The responding firms represented a broad cross section of industries and varied substantially in their amount of capital under management, typical investment longevity, experience, and number of employed professionals. Table 1 provides the data regarding these criteria.

First, the venture capital firms managed an aggregate total of \$5.4 billion of venture capital funds, representing over 20% of the total funds for the industry as of year end 1986. The firm with the most capital under management had \$400.0 million while the firm with the least managed capital had only \$1.5 million. Respondents reported that the average number of years that typical investments were held was 5.6 years, while the average venture capital experience of their firms was 11.0 years. Sixty percent of the responding firms had been in the venture capital business less than 10 years; only six of the responding firms possessed over 20 years of experience. Finally, the responding firms had a mean number of mid-level to senior level professionals of approximately eight and the mean ratio of senior level to lower level professionals within the firms was 1.60 to 1.00.

Statistics on Type of Venture Reported

The ventures in the completed questionnaires also exhibited broad geographic dispersion. California, the Midatlantic states, and the Northeast were the most frequently invested locations with 19, 16, and 12 venture investments.

In addition to broad geographic dispersion, we were primarily interested in ventures that occurred in three popular, but very different, industries for new ventures: computers and related products, medical and health care, and specialty retailing. The responses were heavily weighted towards involvement in the computers and related products industry, which totalled 41 of the 62 sample ventures. This result is not surprising given that approximately 70% of all ventures occur in the high technology category (*Venture Capital Journal* 1984). Eleven reported ventures were in medical and health care and seven in specialty retailing.

TABLE 2 Characteristics of Venture Reported

	Mean	Standard Deviation
<i>Product/service in relation to market</i>		
Innovation demonstrated by product/service	3.87	1.03
Clear market acceptance of product/service	3.47	1.13
Prospects for insulation from competition	3.14	0.97
Benefits of patents, secrets and production methods	3.14	1.26
Market competition at end of second year	3.02	0.85
Anticipated market competition at entry	3.02	0.86
<i>Management team characteristics</i>		
Knowledge of product/service	4.19	0.78
Capacity for sustained effort	3.94	0.93
Knowledge of market for product/service	3.83	1.03
Degree of experience within market for product/service	3.69	1.03
Degree of previous leadership experience	3.35	1.09

Most venture investments were in products rather than services (44 product ventures versus 18 service ventures). Forty-seven investments were made at the introductory stage, versus 12 and 3 in the growth and mature/decline stages, respectively. This bias toward earlier stage investment in the ventures is consistent with the findings of Timmons and Bygrave (1986), which showed that venture capitalists' investments in highly innovative technological ventures are made at early stages of development by a ratio of 2 : 1.

Statistics on Characteristics of Venture Reported

The respondents were asked to identify characteristics of the ventures regarding the product or service in relation to the market and the management team. Previous research has indicated that these criteria are important to the success of a venture.

Table 2 indicates that the mean response to each question regarding the product or service in relation to the market was at least average, which was represented by a value of three. The degree of innovation demonstrated by the product/service was significantly above average. Clear market acceptance and prospects for insulation from competition, both identified by Macmillan, Zemmann and SubbaNarasimha (1986) as important criteria to a venture, were slightly higher than average. The level of competition, identified by Roure and Maidique (1986) as a key predictor of success, at entry and after two years, was average.

For each question regarding management team characteristics, Table 2 indicates that the venture capitalists generally thought that management was above average, which was represented by a value of three. Both knowledge of product or service and capacity for sustained effort, established by MacMillan, Siegel and SubbaNarasimha as important criteria to venture capitalists, were significantly above average. Also significantly above average was management's knowledge of the market for the product or service.

We asked the respondents to rate the performance of the venture, compared to their expectations on four performance criteria, performance being rated on a 1 to 5 scale. A rating of 1 represented performance as being far below expectations, a rating of 3 represented achieving expectations, and a rating of 5 represented far above expectations. Respondents

TABLE 3 Performance of Ventures

	Mean	Standard Deviation
Market share	2.66	0.86
Return on investment	2.50	0.92
Sales volume	2.49	1.07
Net profits	2.26	0.93

were requested to present the ventures' success based on four criteria (sales volume, market share, net profits, and return on investment) in order to avoid arguments regarding which criteria are appropriate in assessing venture performance. The results in Table 3 indicate that the reported ventures achieved below venture capitalist expectations on every one of these four evaluation criteria. Since venture capital firms do not have a high proportion of very successful ventures, these generally mediocre results, for all four criteria, indicate that there was not a bias toward reporting only successful ventures.

Venture Capitalist Involvement

In evaluating the extent to which venture capitalists participated in the activities of the ventures, the respondents were asked to identify, for 20 selected activities:

1. Their degree of involvement (in terms of their contribution to the activity compared to the contribution of the venture team),
2. The venture stage during which involvement occurred,
3. The degree they would change their involvement if they could, and
4. The professional level within the venture capital firm from which involvement occurred.

The stage during which involvement in the activities occurred was used as a control. Four stage criteria were included: seed stage, start up stage, growth stage, and later stage. There was little variance by stage for any of the 20 activities.

Similar results were obtained when controlled for the professional level within the firm at which involvement occurred: mid-level/associate only, mid-level/associate mostly, equal amounts, senior level/partner mostly, senior level/partner only. For all 20 activities, the average involvement was mostly at the senior-level/partner level. This finding supports Gorman and Sahlman's (1986) results that, within a venture capital firm, senior members are the professionals primarily involved in the post-funding management of the venture, and suggests that the mid-level professionals are generally more involved in the pre-funding process. Interviews we have held with venture capitalists support this suggestion.

Of the six most substantial involvement activities, serving as a sounding board to the entrepreneur team had the highest rating for involvement in the ventures (Table 4). This result supports Timmons' (1987) suggestion that one of the most important contributions of a venture capitalist is to act as an advisor. Gorman and Sahlman also suggest that serving as the entrepreneur's confidant is important.

Four of the next five most involved activities are financially oriented: obtaining alternative sources of equity financing, interfacing with the investor group, monitoring financial performance and obtaining alternative sources of debt financing. The high degree of involvement in the financial aspects of the venture is not surprising, and supports research

TABLE 4 Activities With Most Venture Capitalist Involvement

	Mean	Standard Deviation
Serving as sounding board to entrepreneur team	3.77	1.19
Obtaining alternative sources of equity financing	3.63	1.23
Interfacing with investor group	3.62	1.14
Monitoring financial performance	3.18	0.88
Monitoring operating performance	2.82	0.98
Obtaining alternative sources of debt financing	2.70	1.58

conducted by Gorman and Sahlman. In their study, helping to obtain additional financing was ranked first in terms of importance to the venture and was performed in 75.0% of the ventures in their sample.

From Table 5 it is evident that the activities in which the venture capitalists demonstrated the least involvement were in the strategic or operations functional areas. This result suggests that either the venture capitalists did not feel that involvement in these activities was important or that due to the continual nature of the involvement in these activities, the venture capitalists may not have been able to devote substantial time to them.

Because of the similarities in those activities in which the venture capitalists were involved the most and the least, we conducted a rotated orthogonal factor analysis to identify any patterns of involvement. Table 6 displays the results of this analysis and identifies four distinct involvement factors.

The first factor loaded on several activities that entailed involvement in development and operations. As a group, involvement in these activities demonstrated the lowest mean involvement score. It appears that the low involvement resulted because these activities require a substantial amount of hands-on participation, performed on a continual basis. Gorman and Sahlman found that venture capitalists spend approximately two hours of direct attention per week in early stage investments. This limited amount of time per investment does not allow for a substantial amount of venture capitalist involvement in activities that require substantial and continual involvement.

The second factor loaded on activities that entail involvement in the management team selection, which includes searching for candidates, interviewing and selecting management, and negotiating employment terms. Involvement in these activities may require a substantial amount of time, but not on a continual basis.

TABLE 5 Activities With Least Venture Capitalist Involvement

	Mean	Standard Deviation
Selecting vendors and equipment	0.60	0.95
Developing production or service techniques	0.71	0.98
Developing actual product or service	0.72	0.83
Soliciting customers or distributors	0.94	0.94
Testing or evaluating marketing plan	1.64	1.13
Formulating marketing plans	1.85	1.19

TABLE 6 Rotated Orthogonal Factor Analysis for Participation in Venture Activities

	Factor 1 Dev. Oper.	Factor 2 Management Selection	Factor 3 Personnel Management	Factor 4 Financial
Selecting vendors and equipment	<u>85</u>	11	- 1	4
Developing production or service techniques	<u>83</u>	26	- 11	9
Developing actual product or service	<u>79</u>	18	- 11	5
Soliciting customers or distributors	<u>77</u>	16	32	1
Formulating marketing plans	<u>74</u>	15	31	21
Testing or evaluating marketing plans	<u>70</u>	- 32	15	18
Interviewing and selecting management team	16	<u>89</u>	24	22
Searching for candidates of management team	11	<u>89</u>	15	9
Negotiating employment terms with candidates	19	<u>86</u>	20	22
Motivating personnel	- 9	21	<u>88</u>	- 2
Serving as sounding board to entrepreneur team	9	7	<u>77</u>	16
Managing crises and problems	21	38	<u>69</u>	16
Obtaining alternative sources of debt financing	4	20	- 7	<u>86</u>
Obtaining alternative sources of equity financing	10	43	25	<u>68</u>
Monitoring financial performance	30	1	39	<u>66</u>
Eigenvalues	3.89	3.00	2.43	1.90

The third factor represents activities that involve dealing with the personnel of the venture. The three activities included in this factor, serving as a sounding board to the entrepreneurial team, motivating personnel, and managing crises and problems, implicitly entail continual involvement throughout the venture, but not necessarily a substantial amount of time commitment.

The last factor represents activities of a financial nature. As a group this factor exhibited the highest mean score. This result may have occurred because these activities are best suited to the expertise of the venture capitalists and/or because involvement in these activities does not require substantial or continual involvement.

Finally we asked venture capitalists to indicate whether they would have liked to change their degree of involvement for each venture activity. Overall the venture capitalists would change their degree of involvement only slightly. The aggregate mean for the 20 activities over the entire sample was 3.12 on a scale where a value of 3 represented the same degree of involvement as actually occurred. Table 7 displays those activities in which the venture capitalists would have most liked to increase their degree of involvement.

The activities in which the venture capitalists would like to have decreased their degree of involvement are presented in Table 8. All these activities call for involvement in ongoing

TABLE 7 Activities in Which Venture Capitalists Would Most Like to Have Increased Their Involvement*

	Mean	Standard Deviation
Formulating business strategy	3.38	0.70
Formulating marketing plan	3.33	0.61
Serving as sounding board to entrepreneurial team	3.28	0.59
Replacing management personnel	3.26	0.72
Monitoring operating performance	3.24	0.60
Interviewing and selecting management team	3.24	0.66

*A score of more than 3 indicates that the respondent would like to have been more involved in the activity.

operations, which would require a significant time commitment from the venture capitalists. By contrast, those activities in which venture capitalists would have preferred to increase their involvement would not require continuous attention.

Inspection of Tables 4 through 8 indicates that in general venture capitalists in the sample had the following "regrets" regarding their level of involvement in specific activities:

1. None of the venture capitalists regretted their level of involvement in any of those activities in which they were highly involved. Furthermore, although they had considerable involvement in serving as a sounding board and in monitoring operating performance, they would have preferred even more involvement.
2. Although the venture capitalists had little involvement in selecting vendors and equipment, in developing either the production techniques or the actual product itself, and in soliciting customers and distributors, they would have preferred even less involvement.
3. The venture capitalists had little involvement in formulating the marketing plan, but they generally wished that they had been more involved. Possibly they identify this activity as important to the venture success and feel that more involvement in this activity would have contributed to a more successful venture.

TABLE 8 Activities in Which Venture Capitalists Would Like to Have Decreased their Involvement*

	Mean	Standard Deviation
Developing production or service techniques	2.86	0.58
Selecting vendors and equipment	2.88	0.57
Developing actual product or service	2.88	0.60
Soliciting customers or distributors	2.89	0.49
Negotiating employment terms with employees	2.93	0.31

*A score of less than 3 indicates that the respondent would have preferred to be less involved in the activity.

CLUSTER ANALYSIS

Given the variety of venture capital firms in the sample and the variety of investments made, it seemed appropriate to perform a cluster analysis based on the degree of involvement in the activities. This cluster analysis was done using the SAS Fastclus procedure. The number of clusters where the cubic clustering criterion showed a change in slope was 3 so the three-cluster solution was selected.

The results of the cluster analysis are reported in Table 9. What is remarkable about these results is that the patterns of involvement fall into three very clear groups based entirely on the degree of involvement the venture capitalists chose to undertake. First is a group of venture capitalists who exhibited very limited involvement compared to their peers—a group we called the *Laissez Faire* group. Second is a group that exhibited moderate involvement which we called the *Moderates* group. Finally, there is a group that consistently and systematically exhibited more involvement, in virtually every activity, than their peers—a group we called the *Close Trackers*. The *Close Trackers* are *very* involved in their investments. Note that the mean scores for nearly all the activities are 3 or more, which means that these venture capitalists felt that they participated at least as much as the entrepreneurial team.

To determine whether there was an underlying reason why clusters based almost entirely on degree of involvement emerged, we analyzed the means of the general venture and firm characteristics for the three clusters. Table 10 displays the statistics for these characteristics. Our tests indicated that there were no significant differences among the cluster means for any of the characteristics. Thus, based on the data collected in this study, there appears to be no evidence that the three clusters of venture capitalist involvement type resulted from any factor other than that the venture capitalists elected to be that type. In support of this contention was the finding that no significant difference emerged among the three clusters concerning the venture capitalists' regrets, i.e., that they would have liked to change their degree of involvement in an activity, for any of the involvement activities.

We then examined the performances of the ventures within each cluster to determine whether ventures funded by one type of venture capitalist performed better than the others. It was evident that there were no significant performance differences among ventures in the three clusters. In particular, the *Close Tracker* group's investments were no more or less successful, on any of the performance criteria, than the other groups, so it appears that this is not a situation in which the *Close Trackers* were venture capitalists who were closely managing ventures in trouble while the *Laissez Faire* groups were venture capitalists who were leaving their highly successful ventures alone. We were also concerned that the lack of performance differences among the clusters might have been due to different performance expectations among the venture capitalists. For instance, if a *Laissez Faire* had low expectations for a venture that consequently was unsuccessful while a *Close Tracker* had high expectations for a venture that consequently was successful, their responses regarding venture performance might have been similar. However, given the insignificant differences among the clusters' venture firm characteristics and given that venture capitalists in general only fund ventures they expect to be successful, we do not feel that variability in expectations among the clusters caused venture performance evaluations to be similar.

In addition we analyzed the frequency distributions within each cluster, to examine the possibility of a bimodal distribution of the performance criteria. We were particularly concerned with the performance of the *Close Trackers*, where venture capitalists might have become significantly involved because: 1) the venture was performing extremely poorly, or 2) they added substantial value thus causing the venture to be highly successful. None of the three groups exhibited bimodal distribution for any of the performance criteria.

TABLE 9 Cluster Results

	Cluster 1 Laissez faire N = 18		Cluster 2 Moderately involved N = 27		Cluster 3 Close trackers N = 17	
	Mean	SD	Mean	SD	Mean	SD
1. Searching for candidates of management team	0.94	0.94	2.70	1.27	4.06	1.20
2. Interviewing and selecting management team	0.78	0.81	2.78	0.75	4.29	1.26
3. Negotiating terms with management candidates	0.72	0.89	2.56	0.97	4.59	1.23
4. Interfacing with investor group	2.82	0.95	3.56	0.80	4.53	1.18
5. Developing professional support group	1.11	1.08	2.22**	1.20	2.60	1.64
6. Obtaining alternative sources of debt financing	2.22*	1.66	2.33	1.34	3.87	1.30
7. Obtaining alternative sources of equity	2.53	1.18	3.73	0.78	3.86	0.96
8. Formulating initial business strategy	1.83	0.92	2.81	0.88	3.44	1.03
9. Developing production or service techniques	0.81**	0.68	1.12	1.05		
10. Selecting vendors and equipment	0.22	0.54	0.59	0.69	1.42	1.33
11. Formulating marketing plans	0.23	0.55	0.59**	0.57	1.00	1.50
12. Testing or evaluating marketing plans	1.00	0.91	1.81	0.68	2.88	1.41
13. Monitoring financial performance	1.33*	1.24	1.63**	0.79	2.00	1.42
14. Monitoring operating performance	1.00	0.68	1.47	1.18		
15. Serving as sounding board to mgmt. team	2.72*	0.89	3.11	0.64	3.76	0.90
16. Motivating personnel	2.16	0.71	2.81	0.78	3.53	1.07
17. Replacing management personnel (if any)	3.17*	1.34	3.77	0.91	4.41	1.12
18. Managing crises and problems	1.39	1.19	2.19**	1.10	2.76	1.09
19. Managing crises and problems	0.94	1.09	3.30**	0.95	3.47	1.74
20. Managing crises and problems	1.44	0.86	3.03**	0.81	3.47	1.01

*All means significantly different for groups 1 and 2 at 0.05 levels except those thus indicated.

**All means significantly different for groups 2 and 3 at 0.05 level except those thus indicated.

TABLE 10 Characteristics for Ventures Reported per Cluster

	Cluster 1 Laissez faire N = 18		Cluster 2 Moderates N = 27		Cluster 3 Close trackers N = 17	
	Mean	SD	Mean	SD	Mean	SD
<i>Product/service</i>						
Innovation demonstrated by product/service	3.67	1.03	4.00	1.00	3.88	1.11
Clear market acceptance of product/service	3.41	1.12	3.50	1.10	3.47	1.23
Insulation from competition	2.72	0.96	3.26	0.86	3.41	1.06
Benefits of patents, secrets and production methods	2.83	1.38	3.33	1.18	3.18	1.29
Market competition at end of second year	3.07	1.00	3.04	0.82	2.93	0.80
Market competition at entry	3.00	0.97	3.07	0.87	2.94	0.75
<i>Management</i>						
Knowledge of product/service	4.22	0.81	4.04	0.85	4.41	0.62
Capacity for sustained effort	4.06	0.94	4.00	0.73	3.88	1.22
Knowledge of market	4.11	1.02	3.59	0.89	3.94	1.20
Experience within market	4.11	0.90	3.37*	0.93	3.76	1.20
Previous leadership experience	3.67	1.03	3.15	0.91	3.35	1.37
<i>Venture capital firm</i>						
Amount of capital under management (millions)	\$59.85	\$74.88	\$119.31*	\$101.20	\$87.07	\$74.16
Years typical investments are held	5.31	1.30	5.81	1.23	5.53	1.19
Years firm has been in venture capital business	9.29	5.40	12.44	7.64	10.24	6.58
Senior-level professionals	4.06	2.84	4.62	2.12	4.93	2.66
Mid-level professionals	2.69	3.11	2.56	2.73	3.46	2.90
Distance of firm from venture	3.00	1.19	2.89	1.15	3.12	1.32
<i>Performance</i>						
Market share	2.75	0.86	2.77	0.86	2.41	0.87
Return on investment	2.43	1.01	2.60	1.08	2.41	1.18
Sales volume	2.62	0.96	2.50	1.03	2.35	0.70
Net profits	2.33	0.98	2.15	0.78	2.35	1.11

*Means are significantly different from cluster 1 at 0.05 level.

This set of results from our data indicates something very interesting. It is clear that from the point of view of investment performance, there is little point in discussing whether venture capitalists in general should be more heavily or less heavily involved in their investments—those that get heavily involved feel that they do as well, or as badly, as those that take a hands off approach. A more relevant issue in need of examination is the opportunity cost of the heavier involvement.

REGRESSION ANALYSIS

The next step was to regress the involvement variables on venture performance variables for each cluster. First, we ran regressions on the performance variables using the four

TABLE 11 Regression Analyses on Performance for Various Groups

	Sales	Market Share	Profits	ROI
<i>Close Tracker group</i>				
Searching for candidates of management team	-0.22	-0.44 ^a	-0.67 ^b	-0.57 ^a
Negotiating employment terms with candidates	+0.33 ^a	+0.41 ^a	+0.71 ^b	+0.76 ^b
Adjusted r^2	0.16	0.27 ^a	0.53 ^b	0.46 ^b
<i>Moderate group</i>				
Formulating business strategy	-0.37 ^a	-0.34 ^b	0.04	-0.56 ^b
Monitoring operations	+0.47 ^a	+0.37 ^c	+0.42 ^a	+0.53 ^b
Searching for candidates of management team	-0.14	-0.11	-0.25 ^a	-0.43 ^b
Adjusted r^2	0.17 ^c	0.17 ^c	0.34 ^b	0.54 ^d
<i>Laissez-Faire group</i>				
Developing professional support group	+0.76 ^d	+0.77 ^d	+0.73 ^d	+0.76 ^d
Soliciting customers and distributors	-0.72 ^b	-0.59 ^b	-0.56 ^a	-0.36 ^c
Managing crises and problems	-0.67 ^b	-0.52 ^b	-0.66 ^a	-0.71 ^b
Adjusted r^2	0.69 ^b	0.74 ^c	0.62 ^b	0.69 ^b

^a $P < 0.05$.^b $P < 0.01$.^c $P < 0.1$ (reported only where $P < 0.05$ for other regressions).^d $P < 0.001$.

identified factors of involvement, but the results indicated that these factors had no significant correlation with performance. These results could have been due to the loss of variance caused by factoring the involvement variables or because involvement in different activities within a factor had opposite associations with performance. Therefore we performed regressions on the performance variables using involvement in specific activities as the independent variables. Earlier, an inspection of the correlation matrix had revealed a potential problem with multicollinearity among the involvement variables, so we ran stepwise regressions to identify the specific activities for inclusion as independent variables in the regression analysis. Given the small sample sizes of the clusters, we were also interested in selecting the most parsimonious set of explanatory variables.

Table 11 reports the results of the final regression analyses, for each of the three clusters. Each will be discussed in turn, but before presenting the results of the analysis, a general observation is appropriate. Cases where venture capitalist involvement in an activity is not significantly associated with performance should not be taken to imply that such involvement has no effect or association—at this stage the research is too exploratory to reach such conclusions. Furthermore, where significant relations are identified, a negative beta for a particular type of involvement does not mean that involvement should be discontinued, nor does a positive beta mean that a major increase in effort on that particular activity is required. Rather, we suggest that those interventions that *do* reveal significant correlations with performance should be given careful consideration. After all, venture capitalists have always felt very time constrained, so any indication that certain types of involvement might be increased or decreased is worthy of considerable attention. In particular, when there are both negative and positive betas in a regression, redeploying involvement from the negative beta activity to the positive beta activity might lead to significant benefits at no additional effort.

One must be sensitive about inferring cause and effect relationships from the positive or negative correlations between involvement in specific activities and venture performance. We suggest possible explanations regarding the identified relationships, based on our data and subsequent interviews we have held with venture capitalists, but do not imply that they are the only explanations.

Close Tracker Group (17 respondents = 27%)

The Close Tracker group had generally consistent betas across all performance measures, moderate and highly significant r^2 values for the profitability measures, and modest but significant r^2 values for the sales volume measures.

Surprisingly, it appears that excessive involvement in the search for venture management by Close Trackers is associated with performance problems. Note from Table 9 that the *average* level of involvement for this group in the search for candidates of the management team is 4.06, indicating that *on average* the respondents felt they spent much more time than the entrepreneurs searching for management. Clearly, those venture capitalists who scored *higher than the average* must have come close to monopolizing the process, implying that the entrepreneur was hardly involved. We suggest that because the entrepreneur was little involved in the search process, lack of compatibility between the entrepreneur and the candidates may be an explanatory factor for the negative association with venture performance. Managers with first class credentials may have been identified, but then a high degree of incompatibility emerged with the entrepreneur, which may have consequently led to friction and performance problems. Several venture capitalists who were interviewed intimated that the compatibility issue was most relevant when the manager searched for was at the Chief Executive Officer or Chief Operating Officer levels. It was implied that compatibility may have been affected by the entrepreneur's resistance to sharing control or responsibility.

Another possible explanation for the negative correlation between venture performance and Close Tracker involvement in searching for management could be because the venture is already performing poorly. Because of the poor performance the venture capitalists may have wanted to either replace management or bring in additional management, possibly due to lack of confidence in the existing management. In the questionnaire one of the activities for involvement was replacing management personnel. However, involvement in this activity did not demonstrate any significant correlation with any measure of performance for the Close Tracker ventures. In fact, replacing management personnel had no significant correlation with any of the performance criteria for any of the three identified clusters. Therefore, it does not appear that a phenomenon was occurring in which venture performance was poor, leading to search for new management candidates.

Particularly interesting is the strong positive correlation, for *all* measures of performance, between performance and the negotiation of employment terms with the venture management. Venture capitalists, with their experience in establishing compensation structures, could have negotiated compensation incentives (i.e., stock options and bonuses) and corresponding performance evaluation criteria. These compensation structures may have significantly motivated the venture management to maximize the venture's performance. It is interesting to note that in general the venture capitalists identified negotiation of employment terms as one activity in which they would like to have decreased their involvement. Clearly they did not recognize the importance of their involvement in this activity.

It appears that those venture capitalists who elect to become highly involved in their venture investments could do well by taking a less dominating role in the search for management

and by transferring this effort to taking a more aggressive role in negotiating the compensation structures and performance requirements of compatible managers once they are found.

Moderate Group (27 respondents = 44%)

r^2 results for the moderate group were best for the profitability measures, in which beta coefficients were generally significant. Although the betas for the sales volume measures were not significant, the signs of the coefficients for the volume measures were generally consistent with those of the profitability measures.

There were two classes of activity that yielded consistently negative betas: involvement in strategy development and search for members of the management. Moreover, there were consistently positive betas for monitoring of operations, suggesting that the moderately involved venture capitalists could benefit their ventures by trading off some of the time devoted to "big picture" issues, specifically executive search and strategy development, for time devoted to the more concrete activity of assessing ongoing operations. Their moderate levels of involvement presumably allow them to be sufficiently knowledgeable about the business to make monitoring of operations a very worthwhile activity.

The consistency with which there are negative associations between the search for management candidates and the performance of the venture, on all performance measures and for both the Moderates and the Close Tracker investors, calls for a major reassessment of how these two types of venture capitalists structure their involvement. The negative correlation with performance of excessive involvement in the management team search, by both Moderates and by Close Trackers was surprising, since the prior studies we have cited have identified helping to select key management team members as a major value-added contribution of the venture capitalist. Venture capitalists add value during the search process because of their broad range of contacts and resources and there appears to be no reason why venture capitalists should not use such resources and contacts to identify candidates. However, as we suggested, the success of the venture is critically dependent on the ability of the entrepreneur and management to work cohesively as a team. So the venture capitalist should not totally dominate the process, but should ensure that the entrepreneur also participates extensively in the search, so that *compatible*, as well as competent, managers are found.

Laissez Faire Group (18 respondents = 29%)

The results for the Laissez Faire group, reported in Table 11, are perhaps the best set of results, in terms of consistency across *all* performance variables. r^2 values are in the range of 0.6 to 0.7, and all beta coefficients are highly significant. What is particularly gratifying is that similar results are obtained for *several* dependent variables, which increases our confidence in these results.

The message from this set of results for those venture capitalists who adopt a laissez faire style in a venture appears to be—stick to it! Whether profits or sales volume are sought, it seems that it is an ineffective practice to maintain a generally hands off attitude and then, with little experience concerning the details of the venture, to intervene in the process by attempting to solve crises and problems or to solicit distributors and customers. Even if poor performance precipitated the high level of these types of interventions, it is clear that involvement in this activity did not turn the situation around.

It appears that the most constructive role for laissez faire venture capitalists is to deploy their limited attention to building strong professional support groups for the venture—a useful, and easily achievable, strategic contribution that can be accomplished with relatively

little effort. This result supports the contention by Davis and Stetson (1987) that one of the most important value-added contributions of a venture capitalist is to develop the right support group for the venture. By building a strong professional support group to assist in the venture, the venture capitalists are to more effectively maintain their laissez faire style.

In general, the regression results indicate that at least some of the regrets that venture capitalists had regarding their level of involvement in specific activities were justified. Clearly those Laissez Faire investors who wished that they had been less involved in soliciting customers and distributors were correct, since their involvement in that activity had a negative correlation with performance. Similarly, the Moderates who regretted not spending more of their effort on monitoring operating performance were correct, since their involvement in that activity had a positive correlation with performance.

Although this study was limited in scope, its findings provide insights into the venture capital interventions in their investments. In particular, it identified three distinct involvement strategies that appear to be deliberately selected by venture capitalists—high, moderate, and hands-off involvement. It appears that each involvement strategy is about equally effective; the mean performance was similar for all three. What may have to be taken into account is the opportunity cost associated with the higher levels of involvement. The three different involvement strategies appear to call for somewhat different efforts when it comes to specific interventions. Those investors who practice higher levels of involvement could possibly benefit by avoiding an overly dominating role in executive searches. Those that elect to maintain a laissez faire attitude could benefit by building the strongest professional support groups for the enterprise.

Clearly the results raise several new questions that should be answered by further research. Why are there different associations between involvement in venture activities and venture performance among the three identified venture capitalist involvement types? What are the specific cause and effect relationships between involvement in specific activities and venture performance? What should the optimal involvement level be for each type of involvement strategy? Can there be a systematic involvement strategy for each type of involvement strategy? For highly successful ventures what is the value of venture capitalist involvement?

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APPENDIX

Industry in Which Venture Occurred? _____

I. For the following activities that occurred during the venture please
indicate the amount of participation that your firm had in the activity.

	0 No participation at all by your firm	1 Much less than entrepreneur team	2 Less than entrepreneur team	3 As much participation as entrepreneur team
1. Searching for candidates of management team?	0	1	2	3
2. Interviewing and selecting management team?	0	1	2	3
3. Negotiating employment terms with candidates?	0	1	2	3
4. Interfacing with investor group?	0	1	2	3
5. Developing professional support group?	0	1	2	3
6. Obtaining alternative sources of debt financing?	0	1	2	3
7. Obtaining alternative sources of equity financing?	0	1	2	3
8. Formulating business strategy?	0	1	2	3
9. Developing actual product or service?	0	1	2	3
10. Developing production or service techniques?	0	1	2	3
11. Selecting vendors and equipment?	0	1	2	3
12. Formulating marketing plans?	0	1	2	3
13. Testing or evaluating marketing plans?	0	1	2	3
14. Soliciting customers or distributors?	0	1	2	3
15. Monitoring financial performance?	0	1	2	3
16. Monitoring operating performance?	0	1	2	3
17. Serving as sounding board to entrepreneur team?	0	1	2	3
18. Motivating personnel?	0	1	2	3
19. Replacing management personnel?	0	1	2	3
20. Managing crises and problems?	0	1	2	3

[illegible]

III. If you could change the amount that your firm had participated in each activity during the venture, how would the new amount of participation compare?			
	1 Much less	2 Less	3 The same amount
1. Searching for candidates of management team?	1	2	3
2. Interviewing and selecting management team?	1	2	3
3. Negotiating employment terms with candidates?	1	2	3
4. Interfacing with investor group?	1	2	3
5. Developing professional support group?	1	2	3
6. Obtaining alternative sources of debt financing?	1	2	3
7. Obtaining alternative sources of equity financing?	1	2	3
8. Formulating business strategy?	1	2	3
9. Developing actual product or service?	1	2	3
10. Developing production or service technique?	1	2	3
11. Selecting vendors and equipment?	1	2	3
12. Formulating marketing plans?	1	2	3
13. Testing or evaluating marketing plans	1	2	3
14. Soliciting customers or distributors?	1	2	3
15. Monitoring financial performance?	1	2	3
16. Monitoring operating performance?	1	2	3
17. Serving as sounding board to entrepreneur team?	1	2	3
18. Motivating personnel?	1	2	3
19. Replacing management personnel?	1	2	3
20. Managing crises and problems?	1	2	3

Regarding the venture please answer the following questions.

Performance of venture	1 Far below expectations	2 Below expectations	3 Achieved the expectations	4 Above expectations	5 Far above expectations
21. Sales volume?	1	2	3	4	5
22. Market share (of served market)?	1	2	3	4	5
23. Net profits?	1	2	3	4	5
24. Return on investment?	1	2	3	4	5
25. Date of initial investment? _____;			Amount? _____		
26. Date of subsequent investments, if any? _____;			Amount? _____		
			Amount? _____		
			Amount? _____		
			Amount? _____		
27. Location of venture? _____			29. Was the venture for a product or service? _____		
28. Distance from the venture?			30. Stage of market development of the product or service category at the initial investment?		
(a) within 15 minutes.			(a) Introductory Stage.		
(b) 15 minutes to one hour.			(b) Growth Stage.		
(c) one hour to two hours.			(c) Mature or Decline Stage.		
(d) two hours to four hours.					
(e) over four hours.					

Product/service and market	1 Far below average	2 Below average	3 Average	4 Above average	5 Far above average
31. Benefit to venture of patents, trade secrets, and production methods?	1	2	3	4	5
32. Prospects for insulation from competition?	1	2	3	4	5
33. Degree of innovation demonstrated by product or service?	1	2	3	4	5
34. Degree that product or service had clear market acceptance?	1	2	3	4	5
35. Anticipated competition in served market at entry?	1	2	3	4	5
36. Degree of competition at end of second year?	1	2	3	4	5

Management team characteristics	1 Far below average	2 Below average	3 Average	4 Above average	5 Far above average
37. Knowledge of product or service?	1	2	3	4	5
38. Knowledge of market for product or service?	1	2	3	4	5
39. Degree of experience within market of product or service?	1	2	3	4	5
40. Degree of previous leadership experience?	1	2	3	4	5
41. Capacity for sustained effort?	1	2	3	4	5

Please answer the following questions regarding your firm.

42. Number of years that your firm has been in venture capital business? _____
43. Amount of capital under management? _____
44. Number of partners within firm? _____ Number of associates? _____
45. Number of years your firm holds a typical investment? _____
46. Your name? _____
47. Title within firm? _____